Named Entity Recognition (NER)

1) Identify Entites
2) Classify Entites

Entites:-
1) Person
2) Location
3) Date
4) Nos
5) Time
6) Money
7) Percent

Paul is going to Bangkok.
Paul → Person
Bangkok → Location

Dataset :- CoNLL NER Dataset

Applications of NER
1) Information Retrival
2) Question Answering
3) Content Classification
4) Recommendation Systems

Key :- [V1, V2, V3]

Entity

Word Embeddings

Vectorization of Text Data

King = [0.1, 0.2, 0.3, 0.4, 0.5, 0.6]

Queen = [0.1, 0.2, 0.3, 0.7, 0.5, 0.4]

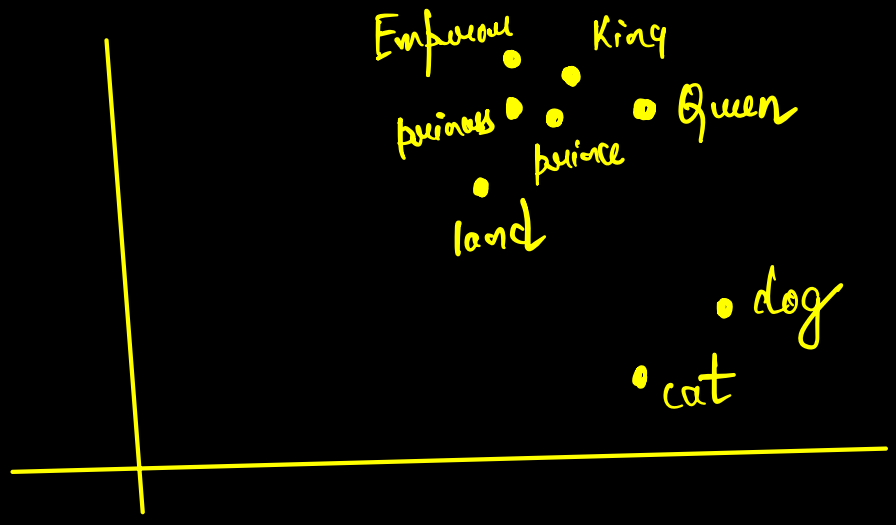
Dimensions => Embedding Model

OpenAI -> text-embed-small (1536)

Gemini -> gemini-embedding-001 (784, 1536, 3072)

3-D Space of Vectors

King = {Emperor, Monarch, power, fame, queen, prince, princess, land}



Vectors

1) Dense → [Non-Zero, Floating 32/64] (256 - 4196)

2) Sparse → multiple zeros (10000 - 50000) mostly